



PLASMA COMPREHENSIVE GENOMIC PROFILING

FINAL REPORT

Patient	Maya Rowan
MRN	MV000017
DOB	11/02/1966
Specimen	Peripheral blood / plasma cfDNA
Collected	05/02/2026 09:16
Report date	05/08/2026

DETECTED ALTERATION

GENE / VARIANT	VARIANT ALLELE FRACTION
ESR1 p.D538G c.1613A>G	3.8%

INTERPRETATION

A pathogenic ESR1 D538G alteration was detected in circulating tumor DNA. ESR1 ligand-binding-domain alterations can emerge in estrogen-receptor-positive metastatic breast cancer after aromatase inhibitor exposure. Interpret this result with the complete clinical history, pathology, imaging, current therapy, and applicable guidelines.

ASSAY

Hybrid-capture next-generation sequencing of plasma cell-free DNA. Reportable alteration classes include selected substitutions, insertions/deletions, rearrangements, and copy-number changes within validated regions.

LIMITATIONS

A negative result does not exclude tumor-associated alterations. Variant allele fraction is influenced by tumor shedding and total cell-free DNA. This report does not independently establish treatment eligibility.

Electronically signed: Priya Shah, PhD, FACMG - Laboratory Director



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TECHNICAL APPENDIX

Specimen quality: PASS. Targeted next-generation sequencing was performed on extracted plasma cell-free DNA. The assay is designed for somatic oncology profiling and is not a germline test.

GENE	VARIANT	CODING	VAF	CLASSIFICATION
ESR1	p.D538G	c.1613A>G	3.8%	Pathogenic

METHOD AND LIMITATIONS

Alterations below the validated limit of detection may not be identified. Copy-number and rearrangement sensitivity varies by locus and specimen tumor fraction. Results should be interpreted by a qualified clinician in the context of other laboratory and clinical information.

Electronically signed: Priya Shah, PhD, FACMG - Laboratory Director